

LUND
UNIVERSITY

Centre for Mathematical Sciences
Mathematics, Faculty of Science

Written examination

The use of electronic devices and additional material (books, notes, etc.) is not permitted during the exam. Justify clearly your answers and please write your anonymous code at the top of each sheet.

1. The *Mangoldt function* Λ is defined by

$$\Lambda(n) = \begin{cases} \log(p) & \text{if } n = p^k, \text{ where } p \text{ is a prime and } k \geq 1, \\ 0 & \text{otherwise.} \end{cases}$$

   (a) Prove that $\log(n) = \sum_{d|n} \Lambda(d)$.

   (b) Prove that $\Lambda(n) = \sum_{d|n} \mu(\frac{n}{d}) \log(d)$, where μ denotes the Möbius function.

2. (a) Show that 2 is a primitive root modulo 13.

   (b) Calculate a table of indices of with respect to the primitive root 2 (modulo 13).

   (c) Find all solutions to the congruence $6x^{10} \equiv 11 \pmod{13}$.

3. Find the infinite continued fraction representation of the number $\frac{1}{2}(3 - \sqrt{7})$. *Hint:* The resulting representation has the form $[a_0; a_1, \overline{a_2, a_3, a_4, a_5}]$.

4. Prove that the odd prime divisors of integers of the form $9^n + 1$, $n \geq 1$, are of the form $p \equiv 1 \pmod{4}$.

5. (a) Prove that the polynomial $x^\ell + 1$ divides $x^{(2k+1)\ell} + 1$, where $k, \ell \geq 1$ are integers. Conclude that if $2^n + 1$ is prime, then n is a power of two.

   (b) Prove that 7 is a primitive root of every prime of the form $p = 2^{4n} + 1$, $n \geq 1$. *Hint:* Determine the quadratic character of 7 modulo p.

6. Let n be a positive integer that is *not* the sum of two squares. Prove that n is *not* the sum of the squares of two *rational* numbers. *Hint:* Argue by contradiction keeping in mind the criterion for a positive integer to be expressible as the sum of two squares of integer numbers.